Data-Driven Strategies for Optimizing Profitability and Operations in a Saree Weaving Business

A Proposal Report for the BDM Capstone Project

Submitted by:

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Declaration Statement

I am working on a Project Title "Data-Driven Strategies for Optimizing Profitability and Operations in a Saree Weaving Business". I extend my appreciation to **Vishwas Maurya Textile**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analysed to assure its reliability.

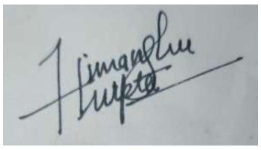
Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfilment in the BS Degree Program offered by IITM Madras. The institution does not endorse any of the claims or comments

Signature of Candidate:



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1. Executive Summary

'Vishwas Maurya Textile' is a small powerloom weaving of saree located at Chhoti Koiran, Lohta, Varanasi. This project focuses on studying, identifying and addressing the operational challenges faced by this business. This business is currently run by brothers 'Mr. Vishal Maurya and Mr. Vishwas Maurya' who primarily focus on B2B markets segment, supplying different sets of designer saree to the Retailer.

The business primarily struggling with the instability issue such as inefficiencies in operations and very nominal profit due to lack of book-keeping and downtime of powerloom machine.

The issues will be tackled by using various analytical methods to examine unorganised data, enabling us to gain insights that will arrive effective strategies.

With the help of recoding transaction data which includes costs, sales prices, and profit margins, to identify patterns and understand the effecting of various factors, this help in setting optimal pricing strategies. Due reduce the chance of downtime of powerloom, which activity has to be record such as operation and idle time, serving period etc.

2. Organisation Background

Business Name: Vishwas Maurya Textile

Address: Chhoti Koiran, Lohta, Varanasi

Owner's Name: Mr. Vishal Maurya and Mr. Vishwas Maurya

Vishwas Maurya Textile is a family-owned powerloom business that has been serving the community in the last 100 years, located in Chhoti Koiran, Lohta, Varanasi. Founded by father of Shri Raj Narayan Maurya who is the father of Mr. Vishal Maurya and Mr. Vishwas Maurya. Initially business establish as a handloom to serve the community and with the effort of Shri Raj Narayan Maurya, business transform into a powerloom. Current owners are polite and approachable brothers; they recently registered their mill under MSME.

3. Problem Statement

1. **Inadequate tracking of production cost and profitability: -**
   The powerloom unit lacks a proper system to track the exact production cost per saree, leading to unclear profit margins. Without knowing the break-even point or accurate cost structure, the owner risks under-pricing the products. This limits financial planning and may cause unintentional losses over time.
2. **Lack of understanding in sales forecasting of Designs/Colour: -**
   The powerloom owner does not maintain records or analyse which saree designs or colours are selling better, especially across seasons. This lack of sales forecasting leads to overproduction or underproduction of certain styles, causing unsold inventory or missed sales opportunities. Without data-driven forecasting, production planning remains reactive rather than strategic.
3. **Lack of Production efficiency metrics: -**
   The powerloom unit does not track key production efficiency metrics such as output per worker, time per saree, or machine downtime. Without these metrics, it’s difficult to identify bottlenecks or areas for improvement in the weaving process. This lack of measurement hinders the owner's ability to optimize operations and scale production effectively.

4. Background of the Problem

Vishwas and his brother face significant challenges related to stagnant profits, downtime of powerloom machine, difficulty in addressing operational cost and arriving appropriate selling price of sarees. The absence of a proper record-keeping system makes it more difficult for them to track monthly income and business operations.

There is uncertainty about the appropriate margin to apply to selling prices in order to achieve the desired monthly income.

5. Problem Solving Approach

To solve the problem of business with a data centric approach, identifying and collecting appropriate data which important in context to our problem by maintaining data attributes such as raw material-wise, colour, style, operational time and many more.

Afterward record and cleaning data to ensure data accuracy and reliability for analysis.

1. Methods Used

Two to three times personal meets and after several talks with owner, its being clearly understandable to explain the requiring solution for business problem.

The business owner gave unorganized data (related to sales, production and costs) which need to structure for further analysis.

Based on the identified problems and project scope, the following data-driven strategies and business methods were adopted to address the operational and profitability challenges: -

1.1 Production Cost Structuring

Analysing and Classifying all fixed and Variable Costs (e.g. electricity, labour, maintenance, raw materials) to understand the break-even point. Identified areas where operational costs could be minimized.

1.2 Profitability Analysis

To understand and identifies the minimum number of sarees that must be sold weekly to avoid losses.

Supplier Cost Negotiation

Exploring opportunities to reduce production costs by discussing better pricing terms with raw material supplier. This indeed comparing alternative vendors and analysing cost-saving opportunities in raw materials without compromising on quality.

2. Intended Data

To thoroughly understand the operational and financial challenges face by the saree weaving unit, I conducted multiple in-depth conversions with the powerloom owner. These discussions provided valuable insights into the business process, limitations and decision-making practices. During the course of these interactions, I gathered key business-related data relevant to the problems identified, this included: -

Production Data: - Weekly quantity of sarees produce, type/design variations and weaving time per saree.

Costing Data: - Fixed costs (machine, rent, electricity, labour) and variable costs (e.g. yarn, dye) per saree.

Sales Data: - Weekly number of sarees sold to the retailer, cost price and selling price.

3. Analysis Tools

1. Microsoft excel: -

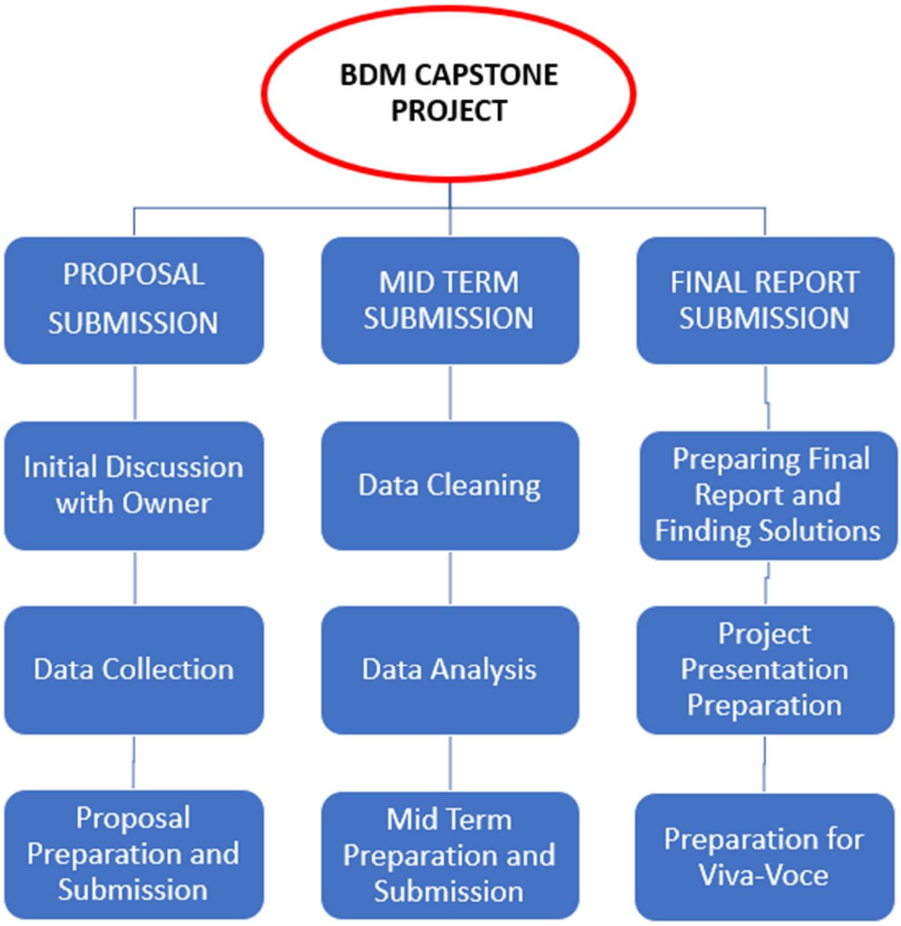
Excel will be used to record, clean, manage and procurement data, helps in visualisation of price trend and track raw materials, inventory level. It also helps in strategic planning through pivot tables and charts which visualize seasonal price variance of raw material and inventory status.

2. Python Tools: -

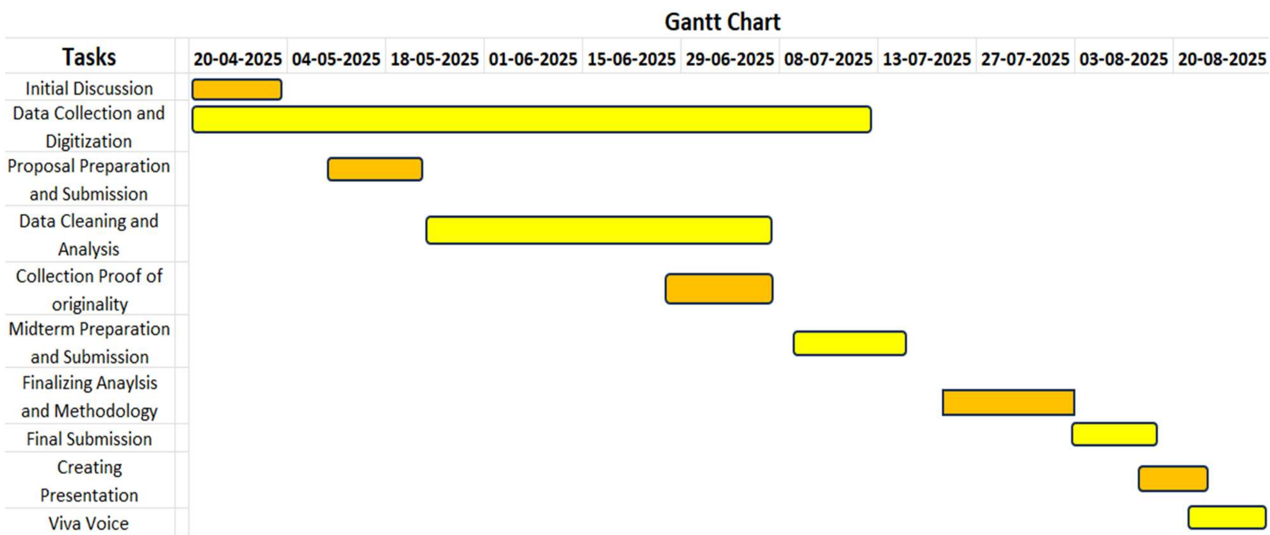
Python, a highly advanced data analysis and forecasting programming language which includes libraries such as Pandas and NumPy which help on exploratory data analysis. Its offer powerful data handling, processing and statistical analysis functions.

6. Expected Timeline

6.1 Work Breakdown Structure



6.2 Gantt Chart



7. Expected Outcome

1. Enhancing business profitability by effectively tracking and establishing pricing strategies that define the optimum profit margin to deal with dealer/retailer.
2. To support the business in establishing inventory management practices by developing data-driven inventory control.
3. Creating appropriate machine maintenance time period which helps in minimizing downtime and cost of repair.
4. Better understanding of demand dynamics related to colour and design, allowing the business to adjust raw material stock level accordingly.